



Thoshith R R 250701797-2025-CSE ▾

T2

Started on	Friday, 21 August 2026, 4:25 PM
State	Finished
Completed on	Friday, 21 August 2026, 4:25 PM
Time taken	35 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00**Problem Statement:**

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

Answer: (penalty regime: 0 %)

```
1  #include <stdio.h>
2
3  int findPair(int arr[], int left, int right, int x) {
4      if (left >= right)
5          return 0;
6
7      int sum = arr[left] + arr[right];
8
9      if (sum == x) {
10         printf("%d\n%d", arr[left], arr[right]);
11         return 1;
12     }
13
14     if (sum > x)
15         return findPair(arr, left, right - 1, x);
16     else
17         return findPair(arr, left + 1, right, x);
18 }
19
20 int main() {
21     int n, x;
22
23     scanf("%d", &n);
24
25     int arr[n];
26     for (int i = 0; i < n; i++)
27         scanf("%d", &arr[i]);
28
29     scanf("%d", &x);
30
31     if (!findPair(arr, 0, n - 1, x))
32         printf("No");
33
34     return 0;
35 }
```

	Input	Expected	Got	
✓	4	4	4	✓
	2	10	10	
	4			
	8			
	10			
	14			
✓	5	No	No	✓
	2			
	4			
	6			
	8			
	10			
	100			

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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